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Artificial Nerve Conduit for Recurrent Laryngeal Nerve Reconstruction in Thyroid Surgery

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Objective: Oncological reconstruction of the recurrent laryngeal nerve (RLN) is sometimes necessary for RLN invaded by thyroid cancer. There have been no case reports of RLN reconstruction using artificial nerve conduits, which are often used for peripheral nerves. In this study, we retrospectively evaluate the feasibility, safety, and efficacy of a collagen conduit with collagen filaments for RLN reconstruction cases at our hospital.

Methods: Artificial nerve conduits were used in seven cases of RLN reconstruction. Two patients had preoperative unilateral vocal cord paralysis with severe vocal cord atrophy, and two had vocal cord paresis without atrophy. The remaining three patients had functional vocal cords before surgery that had to be resected via surgery due to thyroid cancer infiltration of the RLN. Reconstruction was performed using RENERVE®, which is a collagen conduit. Voice examination and laryngeal endoscopy were performed 1, 3, and 12 months after surgery.

Results: There was no improvement in the phonetics of the two patients with vocal cord atrophy before surgery. In the remaining five cases, three with functional vocal cords improved to preoperative values, and two with vocal cord paresis improved to greater than preoperative values.

Conclusion: We report the first case series using an artificial nerve conduit for human RLN reconstruction. In cases of RLN resection when the patient has good voice quality pre-operatively, reconstruction of the RLN using an artificial nerve may be a favorable option in cases where direct anastomosis or ansa cervicalis to RLN anastomosis cannot be performed.

Key Words: artificial nerve conduit, recurrent laryngeal nerve reconstruction, thyroid cancer.

Level of Evidence: 4

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INTRODUCTION

Recurrent laryngeal nerve (RLN) injury impairs vocal cord mobility, resulting in incomplete glottic closure, potentially causing hoarseness, dysphagia, and sometimes aspiration pneumonia, decreasing the patient's quality of life. RLN injury has several causes, including thyroid cancer surgery.[1–4] If thyroid cancer or metastatic lymph nodes directly invade the RLN, it may be resected for curative oncological purposes. In such cases, immediate intraoperative reconstructive procedures such as direct anastomosis (DA) and ansa cervicalis to RLN anastomosis (ARA) may improve speech function and prevent aspiration compared to delayed repair procedures such as injection laryngoplasty and medialization thyroplasty. From this perspective, immediate intraoperative repair of RLN has more advantages than disadvantages and[5,6] should be performed whenever possible.[1,2,7,8]

For peripheral nerve injury reconstruction, DA is optimal for functional results; however, it is impossible with large gaps. Autologous nerve grafts are used as scaffolds to supply nutrients. However, nerve transplants can cause sensory impairments in the original supply area. Therefore, artificial nerve conduits appear to correct defects and restore a similar good function as autologous nerve reconstruction.[9–12] A bioabsorbable collagen conduit filled with collagen filaments (RENERVE®) is currently one of the approved artificial nerve conduits in Japan and is mainly used for peripheral nerve reconstruction after traumatic nerve injury.[9] However, there are no reports on its application in human RLN reconstruction and regeneration. Here, we report our retrospective study findings on the feasibility, safety, and efficacy of a collagen conduit with collagen filaments for RLN reconstruction.

MATERIALS AND METHODS

Filling of Collagen Conduits with Collagen Filaments and Ethical Approval

RENERVE® (Nipro Corporation, Osaka, Japan) comprises a collagen conduit and filaments. Raw collagen is an enzyme-solubilized collagen consisting of a mixture of collagen types I and III. The collagen tube was made of winding threads of solidified collagen in a congealed liquid around the core material. The filaments were made of bundled similar threads. Each of the

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conduits and filaments was provided the necessary in vivo degradation resistance through thermal dehydration and cross-linking treatment. RENERVE® was sterilized by gamma-ray irradiation. RENERVE® was approved by the Japanese Ministry of Health, Labor, and Welfare in 2017 and has been commercially available since 2017.

This study was approved by the ethical committee of Keiyukai Sapporo Hospital (No. 2019–6), and written informed consent was obtained from all patients.

Patients

Between April 2019 and January 2021, seven patients (four men and three women with an age range of 39–82 years) underwent immediate RLN reconstruction with collagen filaments-filled collagen conduits at Keiyukai Sapporo Hospital (Tables I, II) because DA was not possible in these cases. All patients underwent total thyroidectomy for papillary thyroid cancer. Two patients (cases 4 and 7) underwent therapeutic bilateral neck dissection in addition to total thyroidectomy. Three patients (cases 2, 5, and 6) required unilateral therapeutic lateral neck dissection in addition to total thyroidectomy. Furthermore, one patient (case 4) had papillary thyroid cancer infiltration of the contralateral RLN and required direct suturing. Two patients required tracheostomies due to bilateral vocal cord paralysis (case 4) and laryngeal edema (case 7). Regarding preoperative vocal cord movement evaluation in two cases (cases 2 and 3), the vocal fold on the affected side was paralyzed with severe vocal cord atrophy. In two cases (cases 1 and 6), the vocal fold on the affected side showed paresis without vocal fold atrophy preoperatively. The remaining patients had no preoperative vocal fold movement disorder.

Conduit Suturing

The RLN was placed in a reconstruction position and its nerve stump was sharply cut if necessary. Next, a conduit larger than the diameter of the defective nerve was chosen. The tube was soaked in saline solution for 3 min or more (for the saline solution to fill the conduit). Thereafter, the conduit was cut to the length of the nerve defect and sized (proximal and distal) for nerve insertion into the tube. The proximal end of the conduit has a hollow space where the nerve can be inserted; however, on the cut side, its inside was filled with collagen filaments; therefore, pushing into the filaments from the cut side of the conduit creates a hollow space distally for the nerve insertion. The tube was then placed in the reconstruction position. An 8–0 nylon horizontal mattress suture was passed through the epineurium of the nerve, and the nerve was inserted into the conduit with the knot lying on the conduit pipe. The same procedure was performed on both sides of the nerve. The suturing was completed after confirming that both nerve stumps were securely inserted into the conduit.

After surgery, the patient was asked to rest the neck and avoid hyperextending or hyperbending it.

Voice Assessment

Voice function was evaluated by analyzing the maximum phonation time (MPT), jitter (%), shimmer (%), and noise-to-harmonic ratio (NHR) pre- and post-operatively. Postoperative examinations were done for 1, 3, and 6 months and 1 and 2 years after surgery. The vocal signal was recorded using a Kaypentax Computerized Speech Lab (CSL) (Model 4500 and 4250 B) supported by a personal computer, including a Shure-Prolog SM48 microphone placed 20 cm from the patient's lips. Voice sample analysis was performed using the MDVP 5105 software (version 3.2), and jitter, shimmer, and NHR were evaluated. All patients underwent video laryngoscopy and strobolaryngoscopy pre- and post-operatively. We compared voice assessments of the patients with 29 cases (DA/ARA group: DA 13 and ARA 16 cases) treated between 2007 and 2018. Welch's *t*-test was used for the comparison of phonetic parameters between the groups, and the significance level was set at $p < 0/05$.

RESULTS

Proximal 2-mattress and distal 2-matrix suturing of the artificial nerves were possible in the seven cases. None of the patients experienced side effects related to the artificial nerve, including infections and foreign body reactions, during or after hospitalization.

Oral intake by the patients was possible on the first postoperative day. Of the two patients (cases 4 and 7) who underwent tracheostomy, one (case 7) had it closed 2 weeks after surgery because of improved laryngeal edema. The patient was discharged in the third postoperative week. However, it was impossible in the other patient (case 4) because of postoperative bilateral RLN paralysis. Therefore, the tracheostomy tube was replaced with a tracheal silicone stent (Tracheal Stoma Retainer®, KOKEN CO., LTD, Tokyo, Japan) on the fourth postoperative day. The patient was discharged in the second postoperative week while maintaining the tracheal stent. The remaining patients were discharged on the seventh postoperative day.

TABLE I.
Patient Characteristics.

Patient	Gender	Age	Clinical TNM	Preoperative vocal cord	Vocal cord atrophy	Operation of thyroid	Lateral neck dissection	Pathological TNM
1	Female	54	T4aN0M0	Paresis (right)	None	Total thyroidectomy	–	T4aN0M0
2	Male	82	T4aN1bM0	Paralysis (right)	Severe	Total thyroidectomy	Right side	T4aN1bM0
3	Male	75	T4aN0M0	Paralysis (left)	Severe	Total thyroidectomy	–	T4aN1aM0
4	Female	45	T3N1bM0	Functional	None	Total thyroidectomy	Bilateral	T4aN1bM0
5	Female	69	T4aN1bM0	Functional	None	Total thyroidectomy	Left side	T4aN1bM0
6	Male	39	T4aN1bM0	Paresis (left)	None	Total thyroidectomy	Right side	T4aN1bM0
7	Male	45	T4aN1bM0	Functional	None	Total thyroidectomy	Right side	T4aN1bM0

Abbreviation: TNM, tumor, node, and metastasis category.

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TABLE II.
Surgical Records.

Patient	NCR side	Opposite side RLN	RLN diameter	Nerve defect	Length of RENERVE	Inner diameter of RENERVE	Tracheostomy
1	Right	OK	2 mm	18 mm	22 mm	2.8 mm	–
2	Right	OK	2 mm	22 mm	30 mm	2.3 mm	–
3	Left	OK	2 mm	20 mm	24 mm	2.8 mm	–
4	Right	Direct anastomosis	2.5 mm	24 mm	30 mm	2.8 mm	Permanent
5	Left	OK	2 mm	16 mm	20 mm	2.8 mm	–
6	Right	OK	2 mm	16 mm	20 mm	2.8 mm	–
7	Right	OK	2 mm	8 mm	15 mm	2.3 mm	Transient

Abbreviations: NCR, nerve conduit reconstruction, RLN, recurrent laryngeal nerve.

Within 3 months following surgery, all patients received radioactive iodine therapy. Postoperatively, all the patients survived without cancer and underwent voice evaluation. Table III presents the preoperative and postoperative examination results (3 months and ≥1 year postoperatively). In two cases of preoperative RLN palsy and vocal cord atrophy (cases 2 and 3), voice quality deteriorated in 3 months and 1 year postoperatively, respectively. Three patients without preoperative RLN palsy improved their voice quality to the level it was before surgery. The other two patients with vocal cord paresis and no vocal cord atrophy had significantly improved voice quality postoperatively compared to that preoperatively. Table IV compares the voice evaluation of the artificial nerve reconstruction group and the DA/ARA group. There was no statistically significant difference between the two groups.

TABLE III.
Preoperative and Postoperative Voice Quality.

Patient	Pre	Post 1 M	Post 3 M	Post 12 M
Maximum phonation time (second)				
1	7.23	6.62	10.05	15.56
2	6.88	8.94	3.87	0.60
3	1.60	4.65	1.91	0.94
4	21.01	7.54	10.71	21.92
5	9.05	11.58	10.89	10.25
6	16.35	14.86	16.21	18.55
7	28.44	14.21	25.22	39.55
Jitter (%)				
1	2.63	4.87	1.61	0.56
2	3.20	7.66	7	20.31
3	9.74	5.98	12.15	8.8
4	1.6	2.91	1.34	0.86
5	0.44	1.06	0.96	0.48
6	0.24	0.49	2.13	0.55
7	1.31	2.87	0.94	0.16
Shimmer (%)				
1	3	6.54	3.7	2.19
2	4.4	26.97	16.53	20.75
3	37.93	12.48	29.29	23.75
4	3.7	6.6	4.26	3.76
5	1.71	5.85	4.14	2.67
6	2.68	3.69	3.59	3.35
7	1.29	4.42	1.39	1.83
NHR (noise to harmonic ratio)				
1	0.12	0.15	0.12	0.13
2	0.18	0.33	0.42	0.66
3	1.23	0.31	0.42	0.59
4	0.11	0.14	0.12	0.14
5	0.12	0.13	0.15	0.14
6	0.13	0.13	0.17	0.11
7	0.15	0.12	0.15	0.1

Abbreviations: Pre, preoperative; Post 12 M, 12 months postoperative.

CASE REPORT

Case 4

A 45-year-old female patient was diagnosed with thyroid cancer. Preoperative cytopathology of the bilateral thyroid gland and neck lymph nodes revealed papillary carcinoma. Preoperative laryngoscopy revealed normal vocal cord function, and intraoperative neural monitoring (IONM) was not used. The patient underwent bilateral neck dissection and total thyroidectomy. Intraoperatively, the tumor had invaded the bilateral RLN. First, we attempted "partial layer resection of the RLN" reported by Kihara et al.[13] under a microscope; however, we could not perform it. Consequently, we resected the bilateral RLNs. DA without tension could be performed on the RLN left side; however, it was impossible on the right side. Thus "RENERVE®" was used as a scaffold on the proximal and distal stamp on the RLN right side (Figure 1). Conduit suturing was performed as previously described. Finally, the patient underwent a tracheostomy.

Oral meal consumption resumed on the first postoperative day. Postoperative laryngoscopy revealed bilateral vocal cord paralysis. The patient had her tracheostomy tube replaced with a tracheal silicone stent on the fourth postoperative day. Afterward, she could speak with her voice; however, with breathy hoarseness. The patient was discharged in the second postoperative week while maintaining the tracheal stent. The patient was treated with radioactive iodine twice: at 3 and 9 months after surgery.

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TABLE IV.
Comparison of Voice Quality Between the Artificial Conduit Group and DA/ARA Group.

	Artificial conduit group		DA/ARA group		*p* value	
	Pre	Post 12 M	Pre	Post 12 M	Pre	Post 12 M
MPT, s	12.94	15.34	15.58	18.33	N.S.	N.S.
Jitter, %	2.74	4.53	1.97	0.82	N.S.	N.S.
Shimmer, %	7.82	8.33	4.08	3.17	N.S.	N.S.
NHR	0.29	0.27	0.14	0.12	N.S.	N.S.

Abbreviations: ARA: ansa cervicalis to RLN anastomosis; DA, direct anastomosis; MPT, maximum phonation time; NHR, noise-to-harmonic ratio; N.S., not significant; Pre, preoperative; Post 12 M, 12 months postoperative; s, second.

The postoperative voice functional examination results are shown in Figure 2. The MPT, jitter, shimmer, and harmonics-to-noise ratio deteriorated 3 months after surgery and improved to the approximate preoperative value 1 year after surgery. Laryngoscopy video data 1 year after surgery are shown in Video S1.

DISCUSSION

This is the first report to examine the clinical application of a collagen conduit for reconstruction and regeneration in a case series of RLN resections.

Peripheral nerve injury is often traumatic, and that of the hand can cause functional impairment. Adequate nerve reconstruction and regeneration are vital and can affect subsequent functions.[4,9] In cranial nerve injury, the facial and RLNs are two of the most commonly injured nerves.[14] In a tumor-infiltrated facial nerve setting, the affected facial nerve portion must be resected, and intraoperative nerve reconstruction and regeneration would lead to subsequent long-term functional and cosmetic results.[14,15] In RLN injury, immediate primary repair may maintain tone and prevent vocal cord atrophy, arytenoid prolapse, and aspiration; therefore, the benefits should outweigh the potential drawbacks.[1,2,7,8]

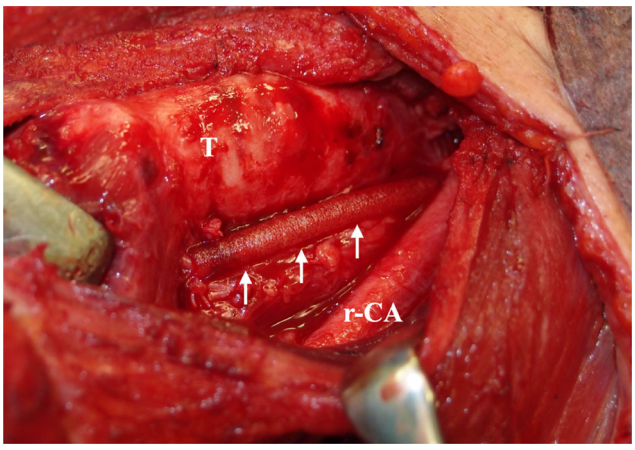


Fig. 1. Intraoperative view of the reconstructed nerve using the artificial conduit nerve. Arrows indicate the artificial conduit nerve; T, trachea; r-CA, right common artery. [Color figure can be viewed in the online issue, which is available at www.laryngoscope.com.]

Regarding nerve reconstruction, when possible, DA of the peripheral and cranial nerves remains the gold standard for reconstruction. If the nerve gap is significantly larger than 5 mm, the direct anastomosis may be suboptimal, as excess tension causes ischemic stress on

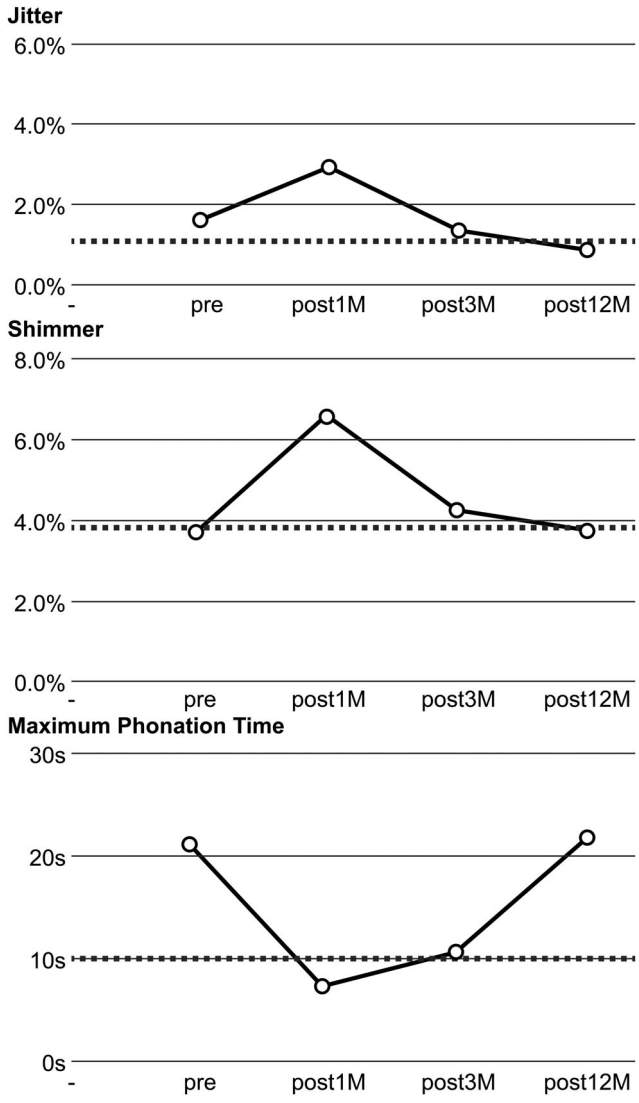


Fig. 2. Voice functional examination results of case 4.

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the nerve and is associated with worse functional outcomes. If the nerve defect is too large for a DA, an autograft is selected; autografts are most suitable for reconstructing peripheral and facial nerves. However, autografts are associated with donor-site morbidity and subsequent sensory deficits. Surgeons have applied a new artificial nerve conduit to overcome these disadvantages and reconstruct peripheral nerves.[4,9,12]

In addition, DA is the preferred option for RLN reconstruction.[16–19] The resected nerve endings are properly sized and carry the most suitable nerve fibers available. In the case of tension, ARA has been used to treat unilateral vocal fold paralysis.[20–22] ARA is as practical as DA for improving phonation in patients who require resection of a unilateral RLN. Ansa cervicalis is a motor nerve that branches to the anterior neck muscle and runs near the RLN; its thickness is suitable for anastomosis of the RLN. No serious functional or cosmetic sequelae occurred at the expense of the nerves. Ansa cervicalis requires only one anastomosis, which can be performed in a simple position near the larynx and is clinically the most effective method for RLN regeneration with satisfactory phonetic results.

Several studies have reported the application of artificial nerve conduits to RLN in animal models.[23–25] In a report comparing artificial nerve conduits and DA in rats, Suzuki et al. reported clearer myelinated fibers and better prevention of laryngeal muscle atrophy in the artificial nerve conduit group than in the DA group.[23] The authors concluded that better nerve regeneration was observed in the artificial nerve conduit group. Şentürk et al. reported that the polyglycolic acid (PGA) conduit could provide a more accurate nerve fiber orientation by creating an isolated environment than the DA in rats.[24] Kanamaru et al. compared the reconstruction with an artificial nerve conduit and an autologous nerve graft after cutting the RLN in beagle dogs[25] and observed functional regeneration in the group reconstructed with an artificial nerve conduit.[25] Similarly, in case 4, artificial nerve reconstruction was performed on the right, whereas DA was performed on the left, and postoperative laryngoscopy showed that the right vocal cord was slightly more mobile than the left one. The authors concluded that an artificial nerve conduit presents a favorable environment for severed nerve regeneration.

The ability of the peripheral nervous system to regenerate,[3] as well as the RLN,[26–28] explains why artificial nerve conduits have shown better results in nerve regeneration than traditional direct anastomosis. Accordingly, animal studies have shown that various neurotrophic factors are expressed during nerve rupture.[3,4] As one of them, the glial cell line-derived neurotrophic factor (GNDF) is up-regulated after nerve injury, stimulates Schwann cell migration and promotes myelination.[3,29–31] It is conceivable that similar changes in neurotrophic factors occur after RLN injury.[28,32–34] By isolating the nerve regeneration process within a conduit, artificial tubes reduce the infiltration of scar tissue from the surrounding environment and concentrate trophic communication between the proximal and distal nerve stumps.[12] Five out of seven patients had good voice examination results after RLN reconstruction, and two of them showed a slight movement of the vocal cord that was reconstructed with the conduit to the median side in synchronization with vocalization. By using artificial nerve conduits, neurotrophic factors can be used intensively, as the filaments inside the conduits can promote regeneration, and the conduits themselves can protect granulation tissue from the surroundings, resulting in good nerve regeneration.

The artificial nerve conduits approved in Japan are the PGA conduits filled with collagen sponge[10] and collagen conduits filled with collagen filaments (RENERVE®).[9] No reports compared these two types; however, collagen conduits have been reported to be suitable for nerve regeneration in animal experiments,[35] and using filaments as the inner material is excellent for nerve regeneration.[36,37] Based on these reports, we decided to use RENERVE®, a collagen conduit filled with collagen filaments.

Studies show that prolonged denervation decreases the degree of functional reinnervation.[38–40] This reduced functional reinnervation may be due to the reduced ability of the Schwann cells to support and interact with regenerating axons in chronically denervated distal nerve stumps. However, many investigations have indicated this is not the case for the larynx.[41–45] Meng et al. showed that the surgical outcome of laryngeal reinnervation 2 years after the injury was better than those of ≥2 years.[41] These findings are supported by the fact that the entire thyroarytenoid muscle did not decrease with long-term denervation in rats and that 6 months after RLN resection, there was no significant difference in the maximal isometric force of the thyroarytenoid muscle between the experimental and control cats.[44] However, there are reports that in patients with unilateral RLN palsy with severe vocal cord atrophy and a large glottal gap, it was difficult to achieve good vocal function with vocal cord nerve regeneration alone.[46,47] The addition of arytenoid adduction to severe breathy hoarseness cases can improve voice quality.[47] In two of our cases, severe atrophy of the vocal cord muscles was observed, suggesting that regenerative ability would be considerably reduced. Furthermore, peripheral nerve regeneration-inducing functions decrease with age in the nerve regeneration induction process of the artificial nerve conduits in mice experiments.[48] A report examining the age of reconstructed patients by ansa cervicalis to RLN anastomosis revealed that laryngeal reinnervation is less effective when the patient's age is >60 years.[49] Both patients were elderly, and their regenerative ability may have decreased due to aging. In such cases, an artificial nerve conduit alone would not result in satisfactory reproduction. In such cases, future improvements are needed using a drug that induces regeneration with the conduit.[50] In this sense, artificial nerve conduit reconstruction is expected to be a platform-like method.

Concerning adverse events, Saeki et al. reported that RENERVE® was used as a peripheral nerve reconstruction material in 47 patients, and a mild infection was experienced in which a causal relationship with the nerve conduit could not be denied in only one case. However, the mild infection improved with antibiotic administration. The authors concluded that no serious adverse

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events were directly associated with the use of an artificial nerve conduit.[9] In a report of 20 cases of peripheral nerve reconstruction using PGA conduits filled with collagen, no adverse events such as kinking or extrusion of artificial nerve conduits, inflammation, or infection were observed.[10] Similarly, in our study, there were no cases of infection, and no other side effects were observed.

This study had some limitations. First is the small number of cases and short observation period. Therefore, further examination is necessary as more cases will be accumulated in the future. Long-term follow-up studies may reveal an increased occurrence of synkinesis as the number of cases increases. The countermeasures should be addressed during follow-up. Second, IONM was used in 4 of the 7 patients (2 patients had preoperative vocal cord paralysis with severe vocal cord atrophy, and 2 had paresis without atrophy). The electrophysiological electromyography response was observed only in one case (vocal cord paresis) but not in the other three cases. Although IONM was not used to guide all RLN management in this study, we suggest the routine use of this tool to examine surgical outcomes and predict voice prognosis after artificial nerve reconstruction. Furthermore, the relationship between the length of the RLN defect and nerve regeneration was not investigated. The size of the defect that indicates artificial nerve reconstruction and whether a good voice can be obtained after reconstruction are essential factors in the clinical determination of the reconstruction method, and the results are awaited. In addition, the kind of voice improvement treatment that would be necessary when severe atrophy of the vocal cords is recognized needs to be examined.

CONCLUSION

A case series of RLN reconstruction using a collagen-based artificial nerve conduit was reported. We reported phonetically favorable and unsatisfactory cases and discussed artificial nerve reconstruction's current limitations and future expectations. In cases of RLN resection when the patient has good voice quality pre-operatively, reconstruction of the RLN using an artificial nerve may be a favorable option in cases where DA or ARA cannot be performed. Furthermore, if a drug for RLN regeneration becomes available, reconstruction using an artificial nerve is expected to become a more phonetically appropriate reconstruction method than the existing reconstruction methods.

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